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PROFILE

MSc student in Artificial Intelligence and Machine Learning at the University of Birmingham (2026–27), now based in the UK. AI security engineer working on the red-teaming and evaluation of LLM and agentic systems, with a defensive background in SOC telemetry and detection engineering. CompTIA CySA+ certified. Current focus: the reliability of the automated judges the jailbreak field uses to score attacks — and what those scores hide. Comfortable owning the full path from telemetry pipeline to trained model to the adversarial evaluation of both.

RESEARCH

Judge-Shaped, Not Harm-Shaped

preprint in preparation | github.com/secslayer/jlens-fuzz

Measurement validity in jailbreak success judges; activation-guided fuzzing.

- Found the standard **hubert233/GPTFuzz** judge — reused across the jailbreak literature without re-validation — inflates attack success by rewarding jailbreak-*shaped* persona/roleplay framing over actual harmful content.
- Built a stricter two-stage replacement (deterministic refusal pre-filter + anti-roleplay LLM-judge rubric); reported honestly that it *still* passed a completion that refuses and offers crisis-support resources — the failure mode survives a targeted fix.
- Reported an honest null: activation-guided span mutation (refusal direction, difference-in-means) shows no consistent ASR advantage over uniform mutation, mechanism confirmed engaged. Smoke-scale (n=5/condition) under a hard free-tier budget, stated plainly.

AJF reproduction

github.com/secslayer/reproductions

- Independent end-to-end reproduction of a 2026 adaptive-jailbreak framework (arXiv:2505.23404) — prompt mutation, binary-tree encoding, dual-cipher output encryption, capability-classification probe — across three local models, plus an ablation and the adaptive defense the original paper names in its limitations but never evaluates.
- Independently replicated the judge-reliability finding: a self-judge was wrong in *both* directions on the same prompts; all verdicts re-checked by hand and the corrections committed alongside the originals.
- Also reproduced **GPTFuzzer** (fuzzing-based automated jailbreak generation) end to end in the same repository.

Agent-protocol authorization audit

deployed multi-agent (A2A) platform | security audit & fix

- Audited authorization on a deployed multi-agent (agent-to-agent) platform against the 14-row protocol threat matrix of Ferrag et al. (*ICT Express* 2025, arXiv:2506.23260), under an evidence bar requiring both a recorded static search *and* a runtime probe before any control is marked absent — no claim of absence by grep alone.
- Found and remediated a missing authorization check on the agent-registry write path; fix verified by a regression test (401 unauthenticated / 403 wrong-permission / 201 authorised).

Agentic-AI attacker benchmark

local LLMs vs a live multi-agent target | private

Does attacker architecture beat model size in agentic-AI exploitation?

- Built an evaluation harness pitting local LLMs (via Ollama) as autonomous attackers against a live, deliberately-vulnerable multi-agent AI application, scored by the target's own event-driven exploit detector as ground truth — no self-judging.
- Reverse-engineered the target's production API (streaming SSE chat, session/CSRF auth, challenge-state detection channel) to run fully automated, reproducible attacks; the client follows session rotation to survive multi-turn runs.
- Implemented and compared three attacker architectures — single-shot, agentic feedback loop, and a multi-agent pipeline (planner/generation/reflection/evolution) — with a rung ladder (drive-workflow / reach-control / bypass) separating capability from execution.
- Honest result: single-shot fails at basic workflow execution; the multi-agent attacker produces coherent social-engineering yet still misses via strategic drift — attacker *architecture*, not model size, dominates. Statistical eval via paired bootstrap CIs.

ENGINEERING PROJECTS

Secure Guard — AI-driven incident response & SOC automation

github.com/secslayer/ai-soc

- Built a telemetry pipeline (Sysmon/Winlogbeat → Logstash → Elasticsearch) normalising 45K+ events, with Python and SQL cleaning to extract analysis-ready threat datasets.
- Trained XGBoost and SVM classifiers with PCA-optimised cross-validation, reaching ~85% accuracy and balanced precision across highly varied malware datasets; Prophet forecasting converted statistical thresholds into concrete response playbooks.

GptGram — production-ready agentic AI platform

github.com/secslayer/Gptgram-production-ready

- Deployed an orchestration backend on AWS — REST APIs and JSON payloads over distributed Celery queues — handling high-concurrency LLM inference for multi-agent reasoning.
- Engineered semantic RAG pipelines on Qdrant, persisting conversational state to S3, under enforced Git/CI-CD and Agile practice.

TECHNICAL SKILLS

- **AI security:** jailbreak evaluation, LLM red-teaming, adversarial ML, judge/eval validity, prompt injection, activation steering and linear probes.
- **Agentic systems:** multi-agent orchestration, RAG pipelines, Qdrant, LangChain, LLM inference at concurrency.
- **Defensive security:** detection engineering, SIEM (ELK, Splunk), incident response, threat hunting, MITRE ATT&CK.
- **ML & data:** PyTorch, scikit-learn, XGBoost, Pandas, NumPy, SQL; core deep-learning architectures (attention, backprop, RNN/LSTM, SVM) derived from scratch by hand.
- **Cloud & engineering:** AWS (EC2, S3), Docker, Git/CI-CD, REST APIs, JSON, Agile/Scrum.

CERTIFICATIONS

- **CompTIA CySA+** (Cybersecurity Analyst, CS0-003) — [verify](#)
- **Mathematics for Machine Learning**, Imperial College London via Coursera — linear algebra, multivariate calculus, PCA — [verify](#)
- **SOC Analyst**, TryHackMe — [certificate](#)

EDUCATION

University of Birmingham — **MSc, Artificial Intelligence & Machine Learning** In progress | expected 2027
TKM Institute of Technology — **B.Tech, Computer Science & Engineering** Graduated 2025 | CGPA 7.56/10.00
Also: Kaggle competitor (ARC Prize 2026, ARC-AGI-3); strong Hack The Box / CTF rankings; TryHackMe and HTB workshops on detection engineering and MITRE ATT&CK.